

Data in Trees

1. Have you completed the Stat 133 Welcome Survey linked on Ed?

Consider the file system below where the root is `/`. The shapes following the names aren't part of the file system; they're pointers used in the questions below.

```

/ △
├── fall
│   ├── math53 ○
│   │   ├── syllabus.pdf
│   │   ├── midterm.pdf
│   │   └── final.pdf
│   ├── stat20
│   │   ├── syllabus.docx ●
│   │   └── labs ▽
│   │       ├── lab1.qmd
│   │       ├── lab2.qmd
│   │       └── lab3.qmd ▲
│   └── history7B □
│       ├── syllabus.pdf
│       ├── essay1.docx ■
│       ├── essay2.docx
│       └── essay3.docx

```

2. What is the absolute path to ■?
3. What is the absolute path to ▽?
4. What is the relative path from □ to ■?
5. What is the relative path from ○ to ■?
6. What is the relative path from ▽ to ●?
7. What is the relative path from □ to ▲?
8. What is the relative path from ▽ to △?

Fill in the blank `_____` with the full command that will generate the output that follows. Let `$` be the character that indicates where you can type at the command line. The working directory at the start of each question is `/`, the root.

9. `$ _____`
`syllabus.pdf essay1.docx essay2.docx essay3.docx`

10. `$ cd fall`
`$ _____`
`fall`

11. In the space below, sketch the file system containing files with the following absolute paths:

```
/yosemite/receipts/gas.jpg  
/yosemite/5400.jpg  
/yosemite/reservations.pdf  
/hawaii/3296.jpg  
/yosemite/receipts/lunch.jpg  
/hawaii/3295.jpg  
/hawaii/3298.jpg  
/yosemite/5399.jpg  
/hawaii/3297.jpg  
/yosemite/receipts/dinner.jpg
```

12. Every directory and file name that you've seen so far in this class has refrained from using special characters. Which characters should you avoid using when naming directories and files?¹

¹You can research the answer to this question online.

Intro to R

13. Have you installed R on your computer? Do you have access to the command line via a shell? This will likely be Terminal on a Mac (which is built-in) or git bash on Windows. See the Tools section of the course website for guidance.

You're encouraged to work on these at your R console to try things out. The next three questions continue with the same R objects. What output will each code block produce?

14. What will the following code produce?

```
a <- 3 + 5  
b <- 4 / 2  
a + b
```

15. What will the following code produce?

```
a <- b - (a / a)
```

16. What will the following code produce?

```
a
```

17. What will the following code produce?

```
log(a)
```

For the following questions, consult the functions help file by using `?` (e.g. `?round`).

18. What modified version of this function would return the value 3.14 instead of 3?

```
round(3.141592)
```

```
[1] 3
```

19. What will the following code produce?

```
ceiling(3.14) + floor(3.14)
```

R on the File System

Assume you start a new R session with your working directory at `~`. Which R functions would you call to do each of the following?

```
~  
├─ Project-1  
│  ├── raw-data-1.txt  
│  ├── processed-data-1.csv  
│  └─ data-processing-1.r  
├─ Project-2  
│  ├── raw-data-2.txt  
│  ├── processed-data-2.csv  
│  └─ data-processing-2.r
```

20. Change your working directory to be `Project-1`?
21. Read the data in `raw-data-1.txt` and save it to the R object named `raw_data_1`.
22. Without changing your working directory, read the data in `raw-data-2.txt` and save it to the R object named `raw_data_2`.
23. You may have noticed that when creating R objects, I avoid certain types of names. What characters are permissible when naming new objects?²

More questions to come...

²You can research the answer to this question online.